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D20A

Roll no - 15

Assignment 1

Explore following 4 websites and answer the questions

- bitcoin.org
- Blockchain.com
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom? = Satoshi Nakamoto
2. Bitcoin transactions are recorded on which data structure? = Blockchain
3. Bitcoin uses which consensus mechanism? = Proof of Work
4. The smallest unit of Bitcoin is called? = Satoshi
5. Bitcoin is a **peer-to-peer** type of what system? = Electronic Cash Payment

Wallets

6. A wallet stores the user's _____ key. = private
7. The recovery phrase used in wallets is called a _____ phrase. = seed

- 8. Hardware wallets store keys _____ (online/offline). = offline
- 9. Bitcoin addresses are derived from _____ keys. = public
- 10. What is required to sign a transaction? (public/private) = private

Security

- 11. Bitcoin prevents double-spending using _____. = consensus
- 12. Bitcoin is secured using _____ cryptography (asymmetric/symmetric). = asymmetric
- 13. To run a Bitcoin node, the software required is Bitcoin _____. = core
- 14. The process of verifying transactions is called _____. = mining
- 15. The Bitcoin network is described as _____ (centralized/decentralized). = decentralized

Mining

- 16. Miners solve _____ puzzles to add blocks. = cryptographic
- 17. Mining rewards a block with newly created _____. = bitcoin
- 18. The block time target for Bitcoin is _____ minutes. = 10
- 19. Mining difficulty adjusts every _____ blocks. = 2016
- 20. Miners compete to find a valid block _____ (hash/signature). = hash

Blockchain Structure

- 21. Each block contains a _____ root. = merkle
- 22. Blocks are linked using the previous block's _____. = hash
- 23. A Bitcoin transaction contains inputs and _____. = outputs
- 24. Unspent transaction outputs are called _____. = UTXOS (Unspent Transaction Output)

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.

bc1qwqdg6squsna38e46795at95yu9atm8azzmyvckulcc7kylcckxswvvzej

16. Record:

- Total BTC received: 3,374,862.67708406 BTC
- Total BTC sent: 3,374,846.83697941 BTC
- Current balance: 13.82753175 BTC

What is the role of a full node?

A full node is a computer that stores a complete copy of the blockchain. It independently verifies all transactions and blocks against Bitcoin's rules, rejects invalid ones, and relays valid ones across the network to maintain decentralization.

List any two developer resources available on bitcoin.org.

The Bitcoin Core Developer Documentation.

The Bitcoin Improvement Proposals (BIPs) index.

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC: [\\$64,558.74](#)
- 24-hour % change: -0.18%
- Market capitalization: \$1.29T
- Circulating supply: [20.06M BTC](#)
- Total/maximum supply: [21.00M BTC](#)

Task C2: Historical Data

- What was the price of Bitcoin one year ago?
One year ago, on August 6, 2025, the price of Bitcoin was **\$113,574.90**.

- Note the highest and lowest prices in the last 30 days.

Looking at the last 30 days (July 7 to August 6, 2026), here are the highest and lowest prices:

- **Highest price:** \$66,160.95 on July 22, 2026
- **Lowest price:** \$61,913.99 on July 9, 2026

Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile?

- **Fixed Supply:** Capped at 21 million coins, meaning sudden demand shifts cause major price swings.
- **Market Sentiment:** Prices are highly reactive to news cycles, hype, and panic.
- **No Traditional Valuation:** Lacks underlying physical assets, dividends, or cash flow.
- **Whale Activity:** Large holders can cause price shocks in the relatively young market.

List two investment risks mentioned.

- **Permanent Loss of Access:** If you lose your digital keys or wallet password, your assets are irretrievable.
- **No Government Insurance:** Unlike traditional bank accounts, crypto lacks FDIC or SIPC protection if an exchange is hacked or goes bankrupt.

♦ Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Blockchain.com provided the clearest understanding because it offers a real-time, visual representation of the network in action. Rather than just reading theoretical definitions, seeing the live ledger operate—much like monitoring a distributed database—made abstract concepts tangible. Tracking live transaction inputs, outputs, and cryptographic hashes in real-time effectively demonstrated how blocks are sequentially chained and verified.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust through decentralization, cryptography, and consensus. Instead of relying on a single central authority (like a bank), a blockchain distributes its digital ledger across a global network of independent computers (nodes). When a transaction occurs, these nodes use complex algorithms to verify its authenticity. Once validated, the transaction is permanently recorded in a "block" and cryptographically linked to the previous one, creating an immutable chain. Because every participant holds a matching copy of the ledger, any attempt to alter data is instantly recognized and rejected by the network, guaranteeing security and transparency without a middleman.